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CHAPTER 1 — INTRODUCTION

1.1 Background

In the telecommunications industry, customer churn has become one of the most critical business challenges. With increasing competition and multiple service providers offering similar pricing plans, customer retention has emerged as a key determinant of long-term profitability. Companies now rely on data-driven insights to predict customer behavior and adopt preventive strategies.

Machine Learning (ML) provides powerful predictive capabilities to identify customers who are likely to discontinue services. By learning patterns from historical customer behavior, ML models enable telecom companies to take timely retention actions, thereby reducing revenue loss.

1.2 Problem Statement

Telecom companies often struggle to forecast which customers are likely to churn. Traditional methods rely on manual assessments or generic rule-based systems that fail to capture complex behavior patterns.

The problem addressed in this project is:

“To develop a machine learning-based predictive model capable of identifying customers at risk of churn based on demographic, usage, billing, and service-related attributes.”

1.3 Objectives

The key objectives are:

- To perform Exploratory Data Analysis (EDA) on telecom churn data.
- To identify and analyze behavior patterns of customers who churn vs. those who stay.
- To preprocess and transform data using encoding, scaling, and imbalance handling.
- To train and evaluate ML models such as Decision Tree and Random Forest.
- To select the best-performing model based on accuracy, F1-score, ROC-AUC, and confusion matrix.
- To deploy the model using a Flask-based web application for real-time prediction.

1.4 Motivation

Customer churn leads to:

- Significant revenue loss
- High re-acquisition costs
- Negative market reputation

Predictive analytics helps organizations intervene early. This motivated the development of a robust, deployable churn prediction system using ML and a user-friendly interface.

1.5 Proposed System

The proposed system consists of:

- Raw dataset ingestion
- Feature engineering and tenure categorization
- One-hot encoding for categorical features
- Imbalanced data handling using **SMOTE-ENN**
- Model training using **Random Forest Classifier**
- Evaluation using standard performance metrics
- Deployment through Flask to enable real-time predictions

1.6 Report Outline

- **Chapter 1:** Introduction
- **Chapter 2:** Methodology
- **Chapter 3:** Results & Discussion
- **Chapter 4:** Conclusion & Future Scope

CHAPTER 2 — METHODOLOGY

2.1 Dataset Description

The dataset used is the **Telco Customer Churn** dataset containing 7,043 records and 21 attributes, including:

- Customer demographics (gender, senior citizen, dependents)
- Account information (contract type, payment method, paperless billing)
- Service usage (internet service, security, TV streaming)
- Billing details (monthly charges, total charges)
- Target variable: **Churn**

2.2 Exploratory Data Analysis (EDA)

EDA was performed to understand:

- Distribution of churn vs non-churn customers
- Numerical variable distributions (monthly charges, tenure, total charges)
- Categorical feature counts
- Correlation heatmap
- Outlier detection
- Tenure group segmentation (1-12, 13-24, ... months)

Important observations:

- Customers with month-to-month contracts churn more.
- High monthly charges increase churn probability.
- Longer tenure customers are significantly less likely to churn.

2.3 Data Preprocessing

Preprocessing steps included:

- Dropping unnecessary columns (e.g., Unnamed: 0)
- Converting TotalCharges to numeric
- Handling missing values
- **Tenure grouping** into 12-month bins:
- 1-12, 13-24, 25-36 ... 61-72
- One-hot encoding for categorical variables
- Balancing the dataset using **SMOTE-ENN**

2.4 Train-Test Split

Data split:

- **Training:** 80%
 - **Testing:** 20%
- Using train_test_split from scikit-learn.

2.5 Model Development

Two models were trained:

- **Decision Tree Classifier**
- **Random Forest Classifier**

Random Forest performed better with:

- Higher ROC-AUC
- Better generalization
- Better handling of categorical interactions

2.6 Model Evaluation

Metrics used:

- Accuracy
- Precision, Recall, F1-score
- ROC-AUC
- Confusion Matrix

Random Forest was selected as the final model based on superior performance.

2.7 Model Saving

The trained model was exported using Pickle:

model.sav

model_columns.pkl

These are reused during deployment to ensure consistent feature ordering.

2.8 Deployment Methodology

A Flask application (app.py) was developed:

- Collects 19 customer attributes from HTML form.
- Converts them into a pandas DataFrame.
- Performs encoding and alignment with model_columns.pkl.
- Predicts churn using the loaded model.
- Displays result + probability on the webpage.

Workflow:

1. User enters customer details
2. Flask prepares input
3. Data is merged with original dataframe
4. Tenure is binned
5. One-hot encoding is applied
6. Missing columns are filled
7. Model predicts churn probability
8. Output is rendered in home.html